

Cloud Computing systems

Lecture 1

Cloud computing

Types of cloud

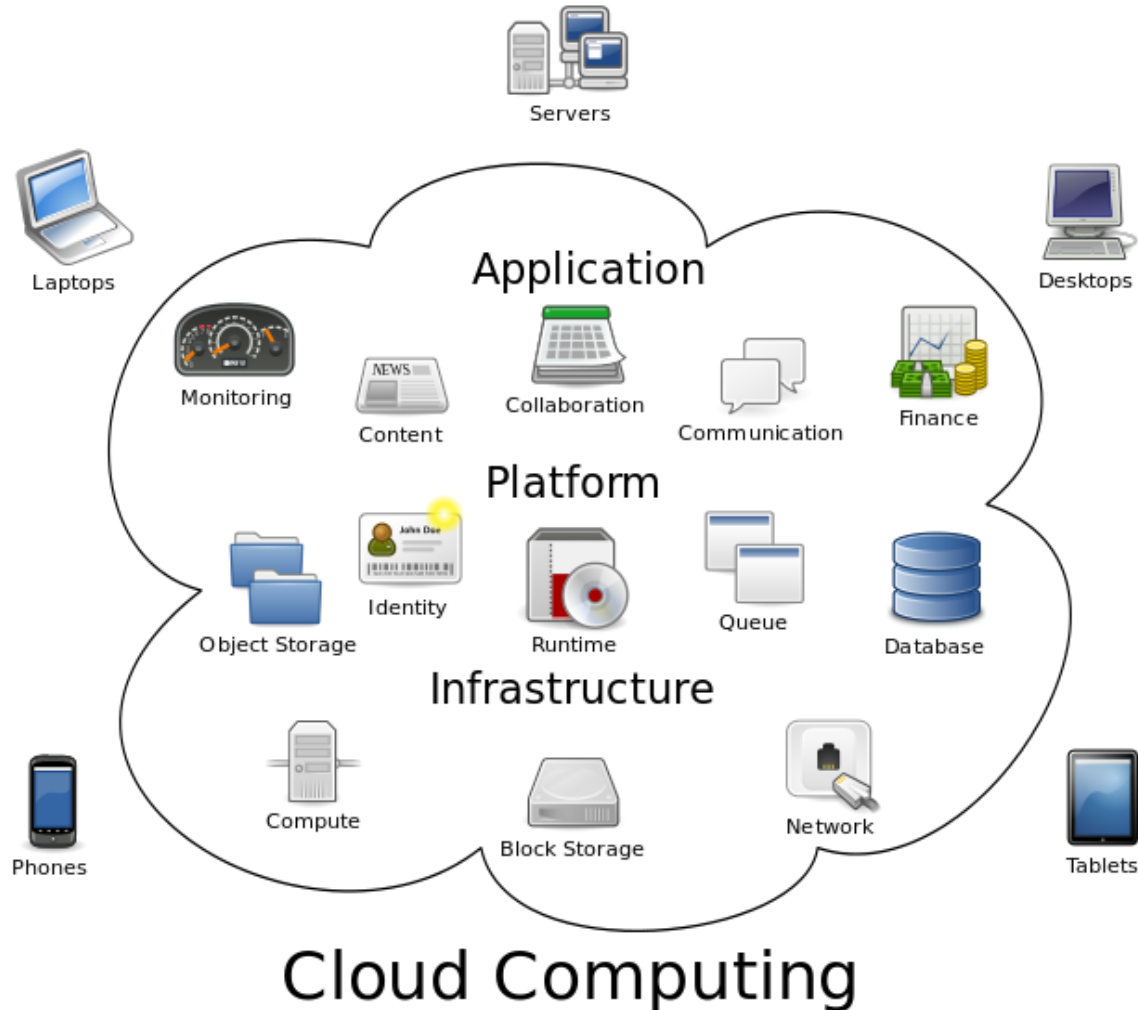
Application of cloud

Cloud computing

- **cloud computing** is a new term in IT domain
- refers to the model of data access, data storage and data processing
- cloud computing model **combines technologies, services and applications into self-service utility** accessible over Internet
- **cloud** part refers to the **abstraction of application details from the user** and virtualization which can be understood as a virtualization of system by pooling and sharing the resources

Cloud computing

The general structure of cloud computing



Cloud computing

Definition

One of the general definitions of cloud computing states that

*“**cloud computing** refers to applications and services that run on a distributed network using **virtualized resources** and **accessed by common Internet protocols** and networking standards. It is distinguished by the notion that **resources are virtual and limitless** and that **details of the physical systems on which software runs are abstracted from the user**”*

Cloud computing

Definition

- National Institute of Standards and Technology (NIST) provides the following definition of cloud computing:

*“**Cloud computing** is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g. networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction”.*

Cloud computing

- main assumptions of the cloud computing are *separation of software architecture from the end user* and *constant access to the application(s) via the Internet connection*
 - these two aspects of the cloud computing allowed to introduce a possibility of usage the same application using various types of devices
- cloud provides various resources through a network
- the customers do not have to install specific software or infrastructure on their devices
- instead can rent them from cloud service vendors

Cloud computing

- in cloud computing the *data is stored in centralized servers* and cached temporarily on clients' devices including desktop computers, notebooks or mobile phones
- this way of resource utilization is the main source of *lower service cost for users* and the main advantage of a cloud
- NIST specifies following essential characteristics of cloud computing:
 - -> ->

Cloud computing

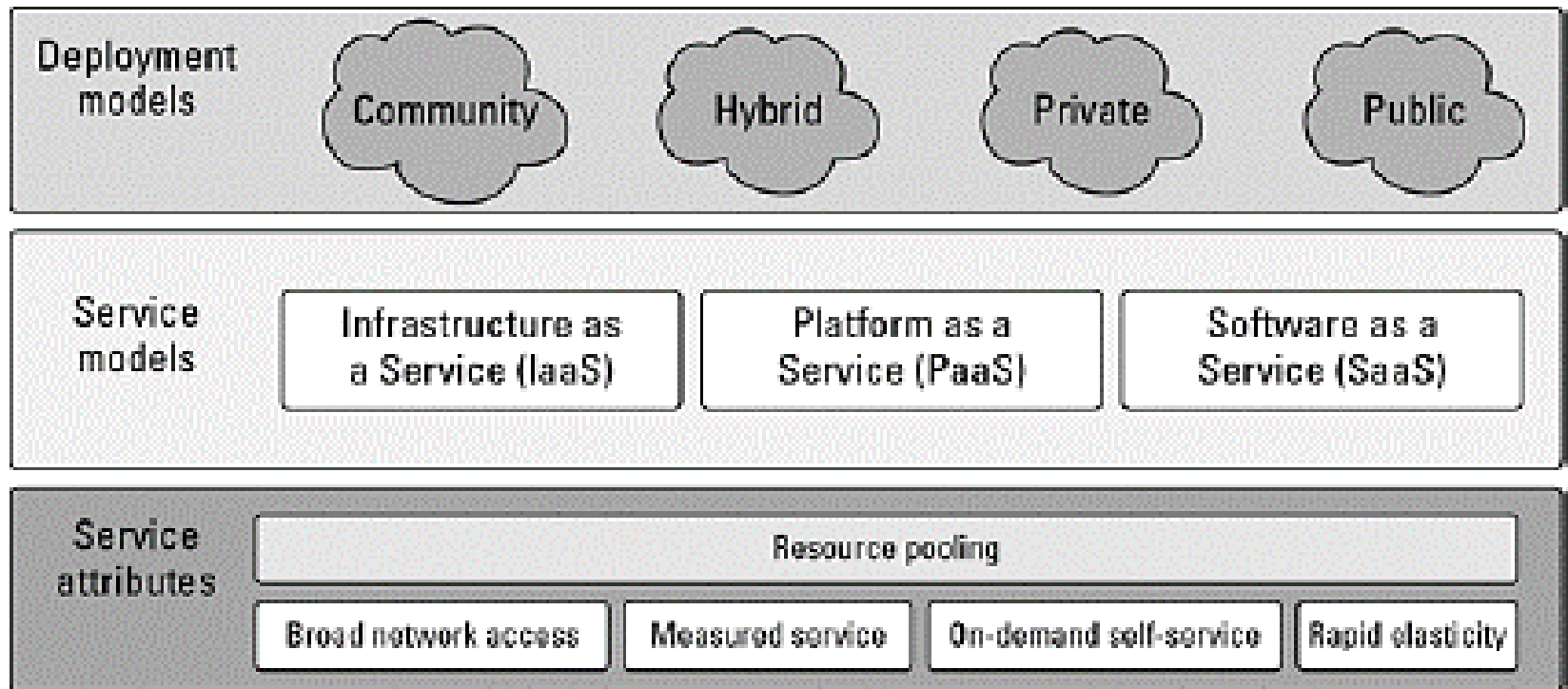
- *on-demand self service* – users can use cloud services like server time, network storage or computations without any human interaction between user and the provider
- *broad network access* – cloud can be accessed by mobile users anywhere they are with any device that can connect to the network laptops, mobile phones, tablets, PCs
- *resource pooling* – in order to meet changing customer demand a large number of resources (e.g. memory, bandwidth, storage) must be available
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Cloud computing

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- *rapid elasticity* – the resources can be allocated or released depending on the needs of a user.
- *measured service* – the resources and services offered by a cloud can be monitored and measured appropriate to type of service (e.g. memory or storage)

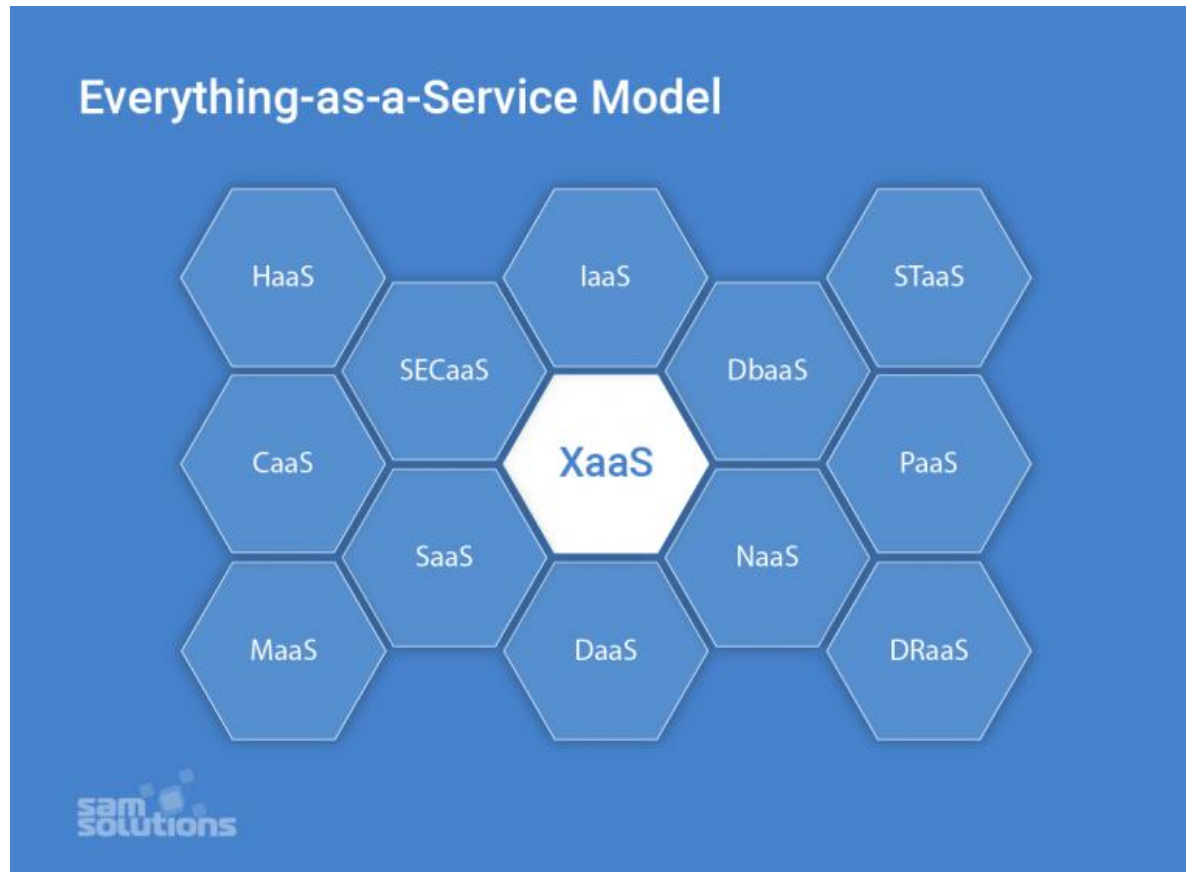
Cloud computing

- in order to provide a precise definition of cloud computing the NIST introduced the division of cloud computing into **deployment models** and **service models**



„Everything” as a Service (EaaS, XaaS, *aaS)

- **Everything-as-a-Service** is a term for services and applications that users can access on the Internet upon request.



„Everything”aaS vs. „Anything”aaS

- **Everything-as-a-Service**
 - designation as-a-Service is associated with various digital components, e.g. data, security, communication, etc.
- **Anything-as-a-Service**
 - another name for XaaS (???)
 - not confined to digital products
 - can get practically everything, from food to medical consultations, without leaving our home or office, by utilizing certain online services
 - hence the “Everything” is in the name.

Applications of cloud computing

You're probably using the cloud at the moment, even if you don't realize it. If you use an online service to send emails, edit documents, watch movies or TV, listen to music, play games or store photos and other files, the cloud service probably works in the background of this service, which makes all this possible.

The first cloud services are only just over a decade old, but now various organizations – from small start-ups through global corporations and government agencies to non-profit organizations – are using this technology for many different reasons.

Applications of cloud computing

- in the late 2000s and into the 2010s, cloud computing was primarily discussed from a hosting position
- companies either migrated to a private cloud, public cloud, or a hybrid cloud model of computing
- today, **hybrid cloud adoption is at 58%** — and growing, and **almost 90% of enterprise organizations are deploying multicloud** solutions
- functional delivery mechanism of cloud technology is no longer about management, and instead revolves around straight-up accessibility, uptime, and availability
- navigating on-premise vs. SaaS vs. PaaS vs. IaaS is almost a moot point

Applications of cloud computing

- reality is that ‘everything’ is serviceable today – that is anything in IT sector can be delivered as a service via the internet

Everything-as-a-Service (XaaS) simply denotes the increasing servitization of technology

Applications of cloud computing

- **Creating native applications for the cloud**
 - Quickly create, deploy and scale applications - web, mobile and API. Take advantage of the advantages of technology and native cloud approaches, such as containers, Kubernetes platform, microservice architecture, API-controlled communication and DevOps infrastructure.
- **Testing and creating applications**
 - Reduce cost and reduce application development time by using infrastructures that can be easily scaled up and down.
- **Analyzing data**
 - Unify your data exchanged between teams, departments and cloud locations. Then use cloud services such as machine learning and artificial intelligence to gain the knowledge you need to make more informed decisions.

Applications of cloud computing

- **Data storage as well as backup and recovery of this data**
 - Protect your data more economically by transferring your data via the Internet to an off-premises cloud storage system that is accessible from any location and using any device.
- **Streaming audio and video**
 - Thanks to global distribution, connect with your recipients anywhere, anytime through any device that supports high definition audio and video.

Applications of cloud computing

- **Embedded intelligence**

- Use intelligent models that can help engage customers and provide valuable detailed information from captured data.

- **Supply of software on demand**

- On-demand software, also known as software as a service (SaaS), allows you to offer customers the latest versions and software updates - whenever and wherever they need it.

DevOps – a relationship of people, processes and technology that enables continuous delivery of value to customers.

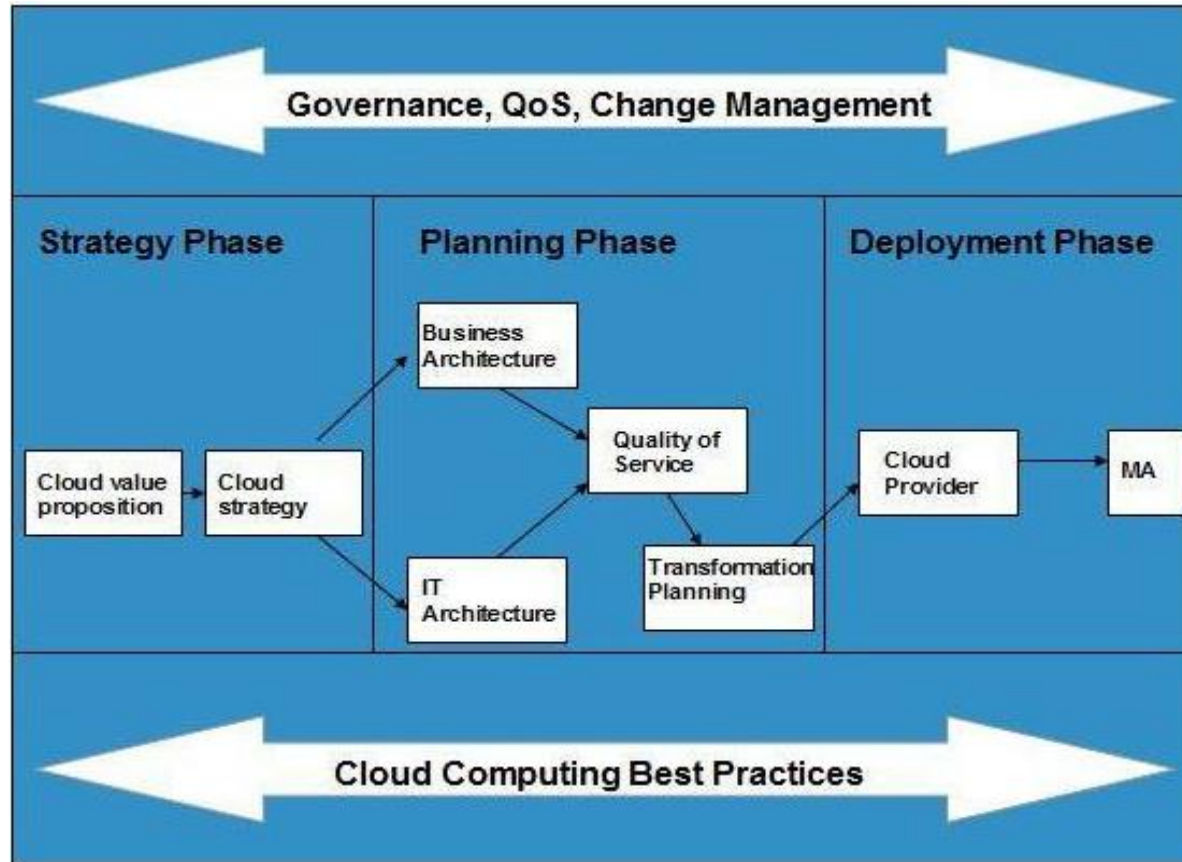
The DevOps methodology brings together programming and operational teams to accelerate software delivery and increase product security and reliability.

Cloud computing planning

- Before implementing cloud computing for your organization, you should consider requirements, especially those related to business operations
- cloud planning process can be divided into three phases:
 - Strategy Phase
 - Planning Phase
 - Implementation Phase

Cloud computing planning

Phases of cloud planning process



Cloud computing planning

Strategy Phase

- in the first planning phase, the value of the cloud offer should be examined
- factors that affect both parties interested in cloud construction and key target issues will be analysed
- key factors are as follows:
 - simplified IT department management (Information Technology)
 - reduction of system operation and maintenance costs
 - an innovative business model
 - low outsourcing costs
 - high quality outsourcing services
- analysis results have a major impact on the definition of planning strategies
- strategy document prepared in accordance with the conditions of cloud computing used in the organization

Cloud computing planning

Planning Phase

- step to determine the plans needed to transform current operations into a cloud computing model
- based on the strategy, threats are analysed that may be caused by adopting a cloud computing model from the perspective of running a business
 - applications that support business management processes and technologies needed to support business applications and data processing systems are defined
- recognizing the requirements for the quality of service development is an important step to the success of cloud computing
- also important non-functional requirements, such as reliability, security, disaster recovery, etc.

Cloud computing planning

Implementation Phase

- includes two stages:
 - choosing a cloud provider
 - provision of maintenance and technical service
- depending on the type of cloud (public, hybrid, private) a cloud service provider is selected
- based on the *Service Level Agreement* (SLA), the level of service that the provider will have to meet is determined
- maintenance and service are provided by the cloud provider
 - such service must ensure appropriate quality of services